Cluster-Based Personalized Mental Wellness Intervention Recommendation System for Students Using Explainable Machine Learning

A mini-project proposal submitted by

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1. Introduction

Mental health plays a vital role in student well-being and academic success. Academic pressure, financial challenges, social expectations, and lifestyle changes have contributed to increasing levels of stress, anxiety, depression, and emotional burnout among students, affecting their performance, productivity, relationships, and overall quality of life.

Advances in Artificial Intelligence (AI) and Machine Learning (ML) enable educational institutions to adopt data-driven approaches for supporting student mental wellness. However, many existing systems focus primarily on identifying mental health concerns and provide limited personalized support. Additionally, their black-box nature often makes recommendations difficult to understand.

This project proposes a Cluster-Based Personalized Mental Wellness Intervention Recommendation System using Explainable Machine Learning. The system will group students with similar mental wellness characteristics into clusters and recommend personalized interventions based on their needs. By incorporating Explainable AI techniques, it will provide transparent and interpretable recommendations, supporting early intervention, personalized wellness planning, and improved student well-being.

2. Need / Motivation

Mental health concerns among students are increasing worldwide, creating a need for effective and accessible support systems. Although many institutions provide counseling services and wellness programs, these services often follow a generalized approach that may not address the unique needs of individual students.

Students facing similar symptoms may have different underlying causes and therefore require different intervention strategies. Furthermore, existing AI-driven systems often lack transparency, making their recommendations difficult to interpret and trust.

The motivation behind this project is to develop a data-driven and personalized solution that can identify different student wellness profiles and provide targeted recommendations. By combining clustering techniques with Explainable AI, the system seeks to improve the effectiveness, transparency, and usability of mental wellness support systems in educational environments.

3. Problem Statement

Students experience mental health challenges due to academic, social, financial, and personal factors. Existing mental wellness systems frequently provide generic recommendations and often fail to account for individual differences. Additionally, many AI-based solutions operate as black-box systems, limiting user trust and understanding.

Therefore, there is a need for an intelligent and interpretable system that can identify distinct student mental wellness profiles and provide personalized intervention recommendations tailored to the needs of each student group.

4. Domain / Application Area

Domain : Healthcare and Education

The proposed project combines concepts from healthcare and education to address student mental wellness through data-driven approaches.

Why this Domain was Selected

Mental health significantly influences student performance, retention, and overall development. Educational institutions increasingly recognize the importance of supporting student well-being but often lack scalable and personalized solutions. The integration of AI and data science provides an opportunity to deliver targeted support while improving decision-making.

Beneficiaries

- Students
- Universities and Colleges
- Academic Counselors

- Mental Health Professionals
- Educational Policymakers
- Student Wellness Centers

5. Business Problems

Lack of Personalized Wellness Support

Most wellness initiatives provide the same recommendations to all students regardless of their individual circumstances.

Delayed Identification of At-Risk Students

Students often seek help only after mental health issues become severe.

Limited Scalability of Counseling Services

Growing student populations can make personalized support difficult to deliver manually.

Lack of Transparency in AI Recommendations

Users may hesitate to trust recommendations when the reasoning behind them is unclear.

Difficulty in Understanding Student Wellness Patterns

Educational institutions often struggle to identify meaningful trends within large student populations.

Inefficient Resource Allocation

Without data-driven insights, institutions may find it challenging to allocate mental health resources effectively.

6. Proposed Methodology

Phase 1: Data Collection

Student mental health datasets will be collected from reliable real-world sources such as Kaggle and the Healthy Minds Study. The datasets may include demographic information, stress levels, sleep patterns, academic pressure, anxiety indicators, and lifestyle factors.

Phase 2: Data Preprocessing

The collected data will be prepared for analysis through:

- Handling missing values
- Removing duplicate records
- Encoding categorical variables
- Feature scaling and normalization
- Exploratory Data Analysis (EDA)

Phase 3: Feature Engineering

Relevant features influencing student mental wellness will be identified and transformed to improve clustering performance.

Phase 4: Student Profiling Using Clustering

Clustering algorithms such as:

- K-Means Clustering
- Hierarchical Clustering

will be used to identify groups of students with similar mental wellness characteristics.

Phase 5: Personalized Recommendation Generation

Each cluster will be analyzed and mapped to appropriate intervention strategies such as:

- Stress management programs
- Sleep improvement plans
- Counseling support
- Mindfulness exercises
- Peer-support activities

- Time-management guidance

Phase 6: Explainable AI Integration

To improve transparency, Explainable AI techniques including:

- SHAP (SHapley Additive Explanations)
- LIME (Local Interpretable Model-Agnostic Explanations)

will be used to explain the factors influencing recommendations.

Phase 7: Web Application Development

The final model will be integrated into a web application developed using Flask and Streamlit, enabling users to access recommendations through an interactive interface.

Phase 8: Testing and Evaluation

The system will be evaluated using:

- Silhouette Score
- Davies-Bouldin Index
- Functional Testing
- Recommendation Validation
- User Acceptance Testing

7. Tools and Technologies

Programming Language

- Python

Frontend

- Streamlit

Backend

- Flask

Database

- MySQL / SQLite

IDE

- Jupyter Notebook
- Visual Studio Code

Libraries

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- SHAP
- LIME
- Plotly

Version Control

- Git and GitHub

8. Product Delivery Mode

Web Application

The final solution will be deployed as a web-based application that enables students to access personalized wellness recommendations and understand the factors influencing them through explainable visualizations.

9. Expected Features

- Student mental wellness assessment
- Cluster-based student profiling
- Personalized intervention recommendations
- Explainable AI insights
- Interactive dashboards and visualizations
- Cluster analysis reports
- User-friendly interface
- Recommendation history tracking

10. Proposed Architecture / Workflow Diagram

Student Dataset

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Data Preprocessing

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Feature Engineering

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Clustering Algorithm

(K-Means / Hierarchical)

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Student Wellness Clusters

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Intervention Recommendation Engine

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Explainable AI

(SHAP / LIME)

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Web Application

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Personalized Wellness Recommendations

11. Expected Outcome

The project will deliver a web-based intelligent recommendation system capable of identifying student mental wellness profiles and providing personalized intervention strategies. By integrating Explainable AI, the system will offer transparent recommendations that improve user trust and understanding. The solution is expected to support preventive mental healthcare, assist educational institutions in decision-making, and contribute to improved student well-being.

12. References

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3. Healthy Minds Network Research Data.
4. Scikit-Learn Documentation.
5. SHAP Documentation.
6. LIME Documentation.
7. Kaggle Student Mental Health Datasets.
8. Research articles on Explainable AI in Healthcare.
9. Research articles on Student Mental Health Analytics.
10. Research papers on Clustering-Based Recommendation Systems.

Student Declaration

We hereby declare that the proposed project work is original and will be carried out under the guidance of the faculty mentor.

Student Signature(s): __________________

Date: __________________

Faculty Guide Signature: __________________